

WED - Jan 22-2025

Topics: chemical rockets mass definitions, single stage & multi-stage rockets, Example, multi-stage rocket optimization.

"Single-stage rockets"

$$\underbrace{M_o}_{\text{initial mass}} = M_L + M_p + M_s$$

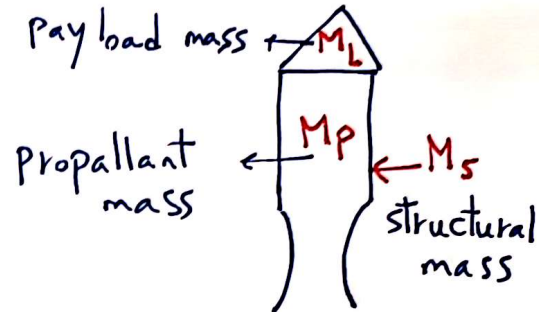
$$\underbrace{M_b}_{\text{Burn-out mass}} = M_o - M_p = M_L + M_s$$

$$\text{Payload ratio: } \lambda = \frac{M_L}{M_o - M_L} \quad \lambda \ll 1$$

$$\text{Structural coefficient: } \epsilon = \frac{M_s}{M_s + M_p} = \frac{M_s}{M_o - M_L} \quad \epsilon < 1$$

$$\text{Mass ratio: } R = \frac{M_o}{M_b} = \frac{1 + \lambda}{\epsilon + \lambda} \quad (\text{It can be proved!})$$

$$\Delta u_{b, \text{imp}} = u_{eq} \ln\left(\frac{M_o}{M_b}\right) = u_{eq} \ln(R) = u_{eq} \ln\left(\frac{1 + \lambda}{\epsilon + \lambda}\right)$$



"Multi-stage rockets"

$$M_{o,1} = M_{L,1} + M_{p,1} + M_{s,1}$$

$$M_{L,1} = M_{o,2}$$

$$\Rightarrow M_{o,1} = M_{o,2} + M_{p,1} + M_{s,1}$$

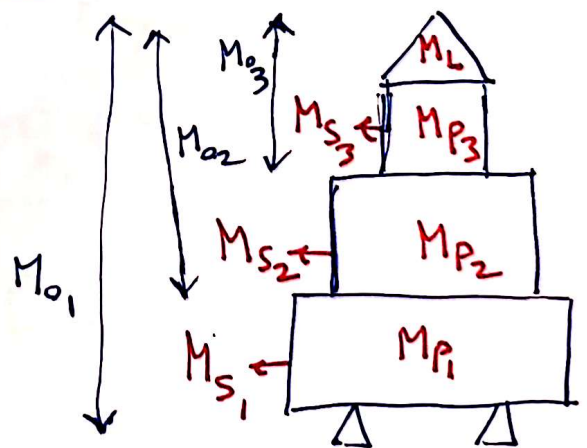


Fig: "3-stage rocket"

$$M_{b_1} = M_{o_1} - M_{p_1} = M_{L_1} + M_{S_1} = M_{o_2} + M_{S_1}$$

$$\lambda_1 = \frac{M_{L_1}}{M_{o_1} - M_{L_1}} = \frac{M_{o_2}}{M_{o_1} - M_{o_2}}, \quad \epsilon_1 = \frac{M_{S_1}}{M_{S_1} + M_{p_1}} = \frac{M_{S_1}}{M_{o_1} - M_{L_1}} = \frac{M_{S_1}}{M_{o_1} - M_{o_2}}$$

$$\Delta u_{b, \text{imp}, 1} = u_{eq_1} \ln R_1 = u_{eq_1} \ln \frac{1 + \lambda_1}{\epsilon_1 + \lambda_1}$$

Similarly, we can write for stages 2 & 3

$$\Delta u_{b, \text{imp}, \text{total}} = u_{eq_1} \ln \frac{1 + \lambda_1}{\epsilon_1 + \lambda_1} + u_{eq_2} \ln \frac{1 + \lambda_2}{\epsilon_2 + \lambda_2} + u_{eq_3} \ln \frac{1 + \lambda_3}{\epsilon_3 + \lambda_3}$$

$$\begin{aligned} \text{total for 3-stage rocket} &= \sum_{i=1}^{n=3} u_{eq_i} \ln \left(\frac{1 + \lambda_i}{\epsilon_i + \lambda_i} \right) \quad \& \quad R_i = \frac{1 + \lambda_i}{\epsilon_i + \lambda_i} \\ &= \sum_{i=1}^n u_{eq_i} \ln R_i \end{aligned}$$

For stage i , we can write:

$$\lambda_i = \frac{M_{o_{i+1}}}{M_{o_i} - M_{o_{i+1}}}, \quad \epsilon_i = \frac{M_{S_i}}{M_{o_i} - M_{o_{i+1}}}$$

$$M_{o_{i+1}} = M_{L_i}$$

$$M_{o_i} = M_{o_{i+1}} + M_{S_i} + M_{p_i}, \quad M_{b_i} = M_{o_{i+1}} + M_{S_i}$$

$$\text{We can show: } \frac{M_{o_1}}{M_L} = \prod_{i=1}^n \left(\frac{1 + \lambda_i}{\lambda_i} \right)$$

↙

$$\frac{M_{o_1}}{M_L} = \left(\frac{1 + \lambda_1}{\lambda_1} \right) \left(\frac{1 + \lambda_2}{\lambda_2} \right) \left(\frac{1 + \lambda_3}{\lambda_3} \right) \quad \text{Multiplication, for 3-stage rocket:}$$

(P-2)

Example: Consider a 3-stages rocket with the following data:

	stage 1	stage 2	stage 3
$M_0(\text{kg})$	2,500,000	580,000	179,000
$M_p(\text{kg})$	1,842,000	373,000	90,000
$M_s(\text{kg})$	96,000	28,000	21,000
$I_{sp}(\text{s})$	370	370	370
$u_{eq}(\frac{\text{m}}{\text{s}})$	3629.7	3629.7	3629.7
$M_L(\text{kg})$	580,000	179,000	68,000
λ	0.3021	0.4464	0.6126
ε	0.05	0.0698	0.1892
$\Delta u_{b,imp}(\frac{\text{m}}{\text{s}})$	4747	3740	2536

$$\Delta u_{b,imp,tot} = 11023$$

$$I_{sp} = \frac{u_{eq}}{g_e} \implies u_{eq} = I_{sp} g_e = 370 \times 9.81 = 3629.7 \frac{\text{m}}{\text{s}}$$

$$\lambda_1 = \frac{M_{02}}{M_{01} - M_{02}} = 0.3021, \lambda_2 = \frac{M_{03}}{M_{02} - M_{03}} = 0.4464, \lambda_3 = \frac{M_{04} = M_L}{M_{03} - M_{04}} = 0.6126$$

(p-3)

$$\varepsilon_1 = \frac{M_{s1}}{M_{o1} - M_{o2}} = 0.05$$

$$\varepsilon_2 = \frac{M_{s2}}{M_{o2} - M_{o3}} = 0.0698$$

$$\varepsilon_3 = \frac{M_{s3}}{M_{o3} - \underbrace{M_{o4}}_{=M_L}} = 0.1892$$

$$\Delta u_{b, \text{imp}_1} = u_{eq1} \ln \frac{1 + \lambda_1}{\varepsilon_1 + \lambda_1} = 4747 \frac{\text{m}}{\text{s}}$$

$$\Delta u_{b, \text{imp}_2} = u_{eq2} \ln \frac{1 + \lambda_2}{\varepsilon_2 + \lambda_2} = 3740 \frac{\text{m}}{\text{s}}$$

$$\Delta u_{b, \text{imp}_3} = u_{eq3} \ln \frac{1 + \lambda_3}{\varepsilon_3 + \lambda_3} = 2536 \frac{\text{m}}{\text{s}}$$

$$\Delta u_{b, \text{imp, tot}} = 4747 + 3740 + 2536 = 11023 \frac{\text{m}}{\text{s}}$$